PRAMUK WIJERATHNE

AI Architect | Solution Architect | Forward Deployed Engineer | Cloud-Native Architecture (Azure/AWS)

Email: pramukha@gmail.com | Phone: +64 21 111 3641 | Location: Wellington, New Zealand

Resident Status: **New Zealand PR** | LinkedIn | GitHub | medium

PROFESSIONAL SUMMARY

AI Solution Architect and GenAI Architect with 20+ years of enterprise software engineering and cloud architecture experience, specialising in designing and delivering AI-enabled enterprise applications. Expert in Generative AI solution architecture, integrating Large Language Models (LLMs) into production-grade systems using Retrieval-Augmented Generation (RAG), vector databases, and agentic AI workflows. Deep expertise in AI orchestration frameworks (LangChain, LangGraph), intelligent automation platforms, and enterprise-grade AI deployment patterns across Azure and AWS. With a strong foundation in .NET, Java, Python, microservices, and distributed systems, I design scalable cloud-native architectures that bridge business strategy with modern AI capabilities across government, banking, and telecom domains in New Zealand and APAC.

TECHNICAL SKILLS

Programming & Frameworks	Java, C#, Python, C | Spring Boot, EJB, .NET 8, ASP.NET Core MVC, Microservices | React.js, HTML5, JavaScript (ES2023), jQuery, Blazor
AI, ML & GenAI	LangChain, LangGraph, Agentic AI, Agentic RAG, Crew AI, MCP (Client/Server), Graph RAG with Neo4j, Light RAG, mem0, Pidentic AI, Crawl4AI , Huggingface, TensorFlow, Distilled Model Finetuning, A2A, MS Power Platform, MS AI foundry and MS AI Framework
Cloud & DevOps	Azure (Functions, API Manager, Entra B2B, Azure AI Search, Container Apps, Key Vault), AWS | Azure DevOps, GitHub Actions, Bamboo, Maven, IaC, Bicep, Terraform | Docker, Kubernetes (AKS, EKS)
Databases & Data	Oracle DB (19c, 21c), PostgreSQL, Azure SQL, Cosmos DB, MongoDB | PL/SQL, MySQL Workbench | Vector DBs: Pinecone, Chroma, Qdrant, PGVector | DB Performance Tuning, Indexing, Partitioning, Migration
Middleware & Servers	IBM Message Broker, IBM MQ, RabbitMQ | Oracle WebLogic, GlassFish, Apache Tomcat, IIS, WSO2
Security	Entra ID, OAuth 2.0, OIDC, JWT, Spring Security, ASP.NET Core Identity | MS Threat Modelling, OWASP Top 10 | Burp Suite, SonarQube
Dev Tools & IDEs	Visual Studio, VS Code (GitHub Copilot), IntelliJ IDEA, JDeveloper, Cursor AI, Claude Code | Linux, Bash | VisualVM
Collaboration & Agile	Git, Bitbucket, SVN, GitLab | Agile (Scrum, XP), TDD (JUnit, NUnit, Mockito), Velocity Tracking

CERTIFICATIONS & AWARDS

- **Winner – Unlocking Productivity Through Connected Health Data** – GovHack 2025 (Infosys) — Dec 2025
- **Agent Skills with Anthropic** – DeepLearning.AI — Feb 2026
- **Agentic AI Certification** – DeepLearning.AI — Oct 2025
- **Building Live Voice Agents with Google's ADK** – DeepLearning.AI — Oct 2025
- **Build AI Apps with MCP Server: Working with Box Files** – DeepLearning.AI — Sep 2025
- **Microsoft Certified: Azure Fundamentals (AZ-900)** – Microsoft — Feb 2023

WORK EXPERIENCE — 20+ YEARS

AI Solutions Architect / Forward Deployed Engineer
Jul 2025 – Present

Inov8 Limited | Wellington, New Zealand

Specialising in enterprise AI solutions and cloud architecture for government agencies.

- Architect multi-agent RAG systems with LangChain, processing 150K+ datasets for multi-label classification across 350+ categories.
- Build locally-hosted AI solutions using Model Context Protocol (MCP) for privacy-sensitive government applications.
- Design Azure-native solutions integrating AI Search, Container Apps, SQL Database, and Key Vault with government-grade security.
- Develop FastAPI microservices and vector search pipelines with PostgreSQL/pgvector for cosine similarity retrieval.
- Create technical proposals for RFPs and FRIs, including solution architectures, security frameworks, and compliance documentation for government clients.
- Define end-to-end architecture blueprints, risk assessments, and delivery roadmaps to support pre-sales and bid responses.
- Collaborate with stakeholders to translate business requirements into technical specifications, ensuring alignment with NZ government security and privacy standards.
- Lead proof-of-concept development and technical demonstrations to validate proposed solutions during procurement processes.

Technologies: Python, FastAPI, LangChain, LangGraph, Azure OpenAI, Azure AI Search, Container Apps, PostgreSQL, pgvector, React, MCP, Docker, Kubernetes.

Senior Software Developer (Technical Lead)
Jan 2023 – May 2025

Datacom | Wellington, New Zealand

Led design, development, integration, and ongoing support of multiple enterprise systems for government and commercial clients including MoJ, Dell Inc., DIA NZ, Downer NZ, DairyNZ, and Spark.

- Delivered full technical support for MoJ's HUB system, collaborating with Fujitsu and MoJ teams; managed CI/CD automation and DB performance tuning. (Team: 8)
- Integrated NZTA safety camera infringement data with MoJ's Court Management System (CMS) and implemented Firearms and Gangs Act legislative reforms. (Team: 3)
- Led development, maintenance, and enhancements of UCIC and CMS systems using Java, EJB, WebLogic, and Oracle DB. (Team: 12)
- Extensively used Azure services including API Manager, Functions, and Entra B2B integration for secure, scalable API-driven services. (Team: 6)
- Contributed middleware application development and bug fixes for Dell Inc.; provided consulting for DairyNZ and Spark projects.
- Delivered business automation and productivity tools using Microsoft Power Platform and SharePoint.

Technologies: C#/.NET 8, Java, Spring Boot, EJB, Oracle WebLogic, GlassFish, Oracle DB 21c, IBM MQ, Bitbucket, Bamboo, Git, SVN, GitLab, Jira, Azure DevOps, Azure API Manager, Azure Functions, Entra B2B, Docker, Kubernetes (AKS, EKS).

Solutions Architect (Lead Service Delivery Manager)
Oct 2021 – Oct 2022

NCS Pte Ltd (Singtel subsidiary) | Singapore

- Led cross-functional teams (QA, BA, developers) delivering quality products for Singapore government agencies.
- Prepared architecture solutions, provided detailed design, and resolved complex coding issues hands-on.
- Created a Learn-All culture across all team functions; managed business, finance, and SLA alignment with the PM.
- Managed application and DB architecture with the Technical Architect; worked within ITIL processes.

Technologies: Java 8, Struts, Spring Boot, Oracle, WebLogic, JavaScript, Git, Jira, Jenkins, Confluence.

Product Development Manager
Oct 2019 – Sep 2021

Assimilated Technologies Pte Ltd | Singapore / *Team of 6 (onshore/offshore)*

- Led design, development, and management of Access and Visitor Management Systems for Resorts World Sentosa, ICA Singapore, banks, and airports.
- Redesigned monolithic platform to event-driven microservices using Azure Functions and Service Bus.
- Developed secure APIs (OAuth 2.0, OpenAPI) integrating with Opera Backend; integrated DBS POS, Alipay, and Opera Payment Interface.

- Implemented OCR and liveness detection using Acuant and Azure APIs for kiosk and iGate systems.
- Oversaw migration of legacy systems to Azure-based microservices, enhancing scalability and performance.

Technologies: C#, .NET Core 3.1/5, ASP.NET MVC, Entity Framework, SQL Server, React.js, SignalR, WebSocket, Azure, Docker, Jenkins.

Technical Lead Developer
Apr 2019 – Aug 2019

Allied Container Group | Singapore / *Team of 4*

- Restructured legacy systems to integrate with 3rd-party systems and implemented microservices for existing monolithic services.

Technologies: Java, C#, Spring Boot, PHP, Yii2, microservices, JavaScript, jQuery.

Technical & Implementation Lead (Payment Expert)
Oct 2014 – Oct 2019

BPC Banking Technologies | Singapore / *Team: 5–8*

Led end-to-end payments architecture and delivery of the SmartVista platform across 10+ APAC banks.

- Designed end-to-end payment architectures covering card issuing, acquiring, clearing, and settlement on the SmartVista platform deployed across Unix/Oracle high-load distributed environments.
- Architected ATM, POS, and Kiosk integration with high-volume, low-latency switch processing on ISO 8583 protocols; supported NDC, DDC, TCP/IP, and X25 interfaces.
- Delivered fraud management workflows with real-time decisioning (allow/decline/block) and instant alerting to issuers, acquirers, and cardholders via SMS and email.
- Owned network development and certification on VISA, Mastercard, UPI, NSS, Bancnet, Diners Club, Europay, Prima, Alto, and Amex (acquiring and issuing).
- Integrated HSMs (SafeNet, Thales) for cryptographic operations and core banking platforms (Finacle, Flexcube, CBS T24) for reconciliation, disputes, and settlement.
- Led SmartVista deployments for HNB Sri Lanka, NEPS Nepal, Sinarmas Indonesia, Warehouse NZ, KBZ Myanmar, UCB, PVB Philippines, HKL, Amret, and CPB Cambodia.
- Conducted architecture governance, design reviews, and mentored engineering teams; coordinated technical implementation across multiple application teams for production cutovers.

Associate Technical Lead
Dec 2013 – Oct 2014

hSenid Software International | Sri Lanka / *Team: 8–10*

- Worked as NoSQL Specialist for Big Data handling; re-designed reporting system to improve database performance.
- Resolved live issues and ongoing developments on WSO2 product stack (Application Server, BPS, ESB, Identity Server, Governance Registry).

Technologies: JavaEE, Spring, Struts2, Hibernate, jQuery, Ajax, MongoDB, MySQL, Drools, WSO2.

Senior Software Engineer
Jun 2008 – Nov 2012

Codegen International | London, UK / Sri Lanka / *Team: 5–6*

- Designed and developed TravelBox travel software for hotel and flight booking engines with SOAP web services.
- Delivered projects for DTW (United Kingdom), Goway Travels (Canada), and ANWB Holidays (Netherlands).
- Built CRM modules, Oracle 11g database designs, payment gateway integrations (Servebase), and web service clients using JAX-WS.

Software Engineer
May 2005 – Jun 2008

Wavenet International (Pvt) Ltd | Sri Lanka

- Designed and developed SMSC, IVR, SIP, and VoIP applications for the Telco industry using Erlang, C, and Java.
- Implemented Media Gateway and Media Server applications on Dialogic NaturalAccess for unicast/multicast voice and video streaming over IP.

Software Engineer
May 2005 – Apr 2006

Lanka Bell (Pvt) Ltd | Sri Lanka

- Developed back-end rating, pre-rating, and billing programs for telecom operator using Pro*C and Oracle PL/SQL.

Software Engineer 2004 – 2005

Celetronix | Sri Lanka

- Redesigned memory test patterns and wrote chip register programs for SDRAM cell repair using Micron redundancy repair algorithms.
- Developed BIOS read/write programs for motherboard testing using Borland C and TASM assembler.

KEY PROJECTS

CAS-Agentic-RAG: AI-Powered Traffic Incident Analysis System | Inov8 / NZTA | Oct 2025

- Built a FastAPI backend with multi-agent pipeline (SimilarityEngine, ContextAnalyzer, FactorPredictor, EvidenceRanker) for structured prediction workflows.
- Implemented RAG using PostgreSQL with pgvector for cosine similarity search over 1536D embeddings from Azure OpenAI.
- Developed a React frontend with real-time analysis dashboards for NZTA traffic incident contributing factor prediction.

AI-Powered Customer Service Chatbot – Public Sector | Personal R&D | 2025

- Designed and built a production-ready multi-agent AI chatbot enabling residents to get instant answers about services, policies, and FAQs via text and voice.
- Implemented Smart Router (hybrid intent classification), RAG Agent (semantic search, re-ranking, multi-query expansion, Map-Reduce summarisation), Planner→Executor→Validator pipeline, Content Moderator, and Urgency Detector.
- Built full document ingestion pipeline (parse→chunk→embed→store) with OCR via Tesseract and Azure Document Intelligence; pgvector semantic search on PostgreSQL.
- Integrated Redis-backed semantic cache with vector similarity search, Azure Speech Services (TTS), and Faster-Whisper (STT) over WebSocket streaming for low-latency voice conversations.
- Implemented RBAC, JWT auth, content moderation, OpenTelemetry tracing with Jaeger, and deployed to Azure Container Apps with automated migrations and health checks.

Resorts World Sentosa – Guest Check-in Cloud Platform | Assimilated Technologies | 2019–2020

- Won the bid by developing MVP; built cloud-native guest check-in system reducing check-in time from ~3 min to ~45 sec.
- Full-stack C#/.NET Core and React portal for kiosk, mobile, and admin UIs with OCR and liveness detection.

SmartVista Payment Solutions – Asia Pacific Banks | BPC Banking Technologies | 2014–2019

- Led SmartVista deployments for 10+ banks across APAC (Sri Lanka, Nepal, Indonesia, Myanmar, Philippines, Cambodia).
- ATM/POS/Kiosk integration, ISO 8583 certification on Visa/Mastercard/UPI networks, fraud management systems.

EDUCATION

Postgraduate Diploma in Business Administration (IT) 2007 – 2009

University of Moratuwa, Sri Lanka

BSc (Special) Engineering – Computer Engineering 1999 – 2004

University of Peradeniya, Sri Lanka | Merit Pass | GPA: 3.45 / 4.2

St. Servatius' College, Matara

Mathematics Country Rank: 169

PROFESSIONAL AFFILIATIONS

- Associate Member – Institute of Engineers Sri Lanka (IESL)
- Member – Institute of Electrical and Electronics Engineers (IEEE)
- Member – Engineering Council Sri Lanka (ECSL)